photographs must be taken in order to be lucky enough eventually to see a nuclear transmutation take place.

In Figure 10 we see a large number of alpha particle tracks, going from right to left. But at one point, an individual diagonal track crosses the main path of the alpha particles, travelling upwards and slanting to the right. If we study the photograph more closely, we can detect a second very dense track, which starts out from the point of origin of the former and runs diagonally to the left sloping slightly downwards. This point is where a nuclear transmutation took place, produced by an alpha particle. These two tracks are: first, the path of the proton knocked out of the nucleus, and, second, the path of the transformed nucleus which received a powerful impact in this process.

These experiments indicate that protons are, in all probability, the fundamental building blocks of atomic nuclei, and this is reminiscent of Prout's hypothesis that all atoms are formed from hydrogen.

Now in 1932 a particle of a previously totally unknown kind was discovered, which can be knocked out cf atomic nuclei by an analogous method. This discovery was the achievement of three scientists, Joliot, Curie and Chadwick, who followed a path first taken by Bothe in Germany. This particle, of practically the same mass as a proton, carries no electric charge, and therefore leaves no visible track in the cloud chamber. It was named *neutron.* The first nuclear reaction in which the emission of a neutron could be observed, was the transmutation of beryllium. Beryllium atoms, mass number 9, atomic number 4 ($_4Be^9$), were bombarded by alpha particles, $_2He^4$; the product was a carbon nucleus of mass number 12 and atomic number 6. The following equations show the masses and charges (mass numbers and atomic numbers) of the particles involved:

$$9 + 4 - 12 = 1 \textit{ and } 4 + 2 - 6 = 0$$